



Assistive Arthritis Device (FDP) 2025-2026

Kushal Dave, Angelina Garcia, Annelise Oka, Alex Ervin, Devansh Patel, Amir Saatchi
EMPOWER Student Association, California Polytechnic State University, San Luis Obispo, CA



Project Description

Who: Christina is the parent of a Cal Poly student and suffers from chronic pain due to osteoarthritis in her right elbow and hand.

What: The assistive device is designed to provide an upward force on her forearm and allow support for everyday movement.

Why: The device strives to reduce the amount of stress imposed on Christina's elbow by counterbalancing the weight of her forearm through the use of torsional springs.

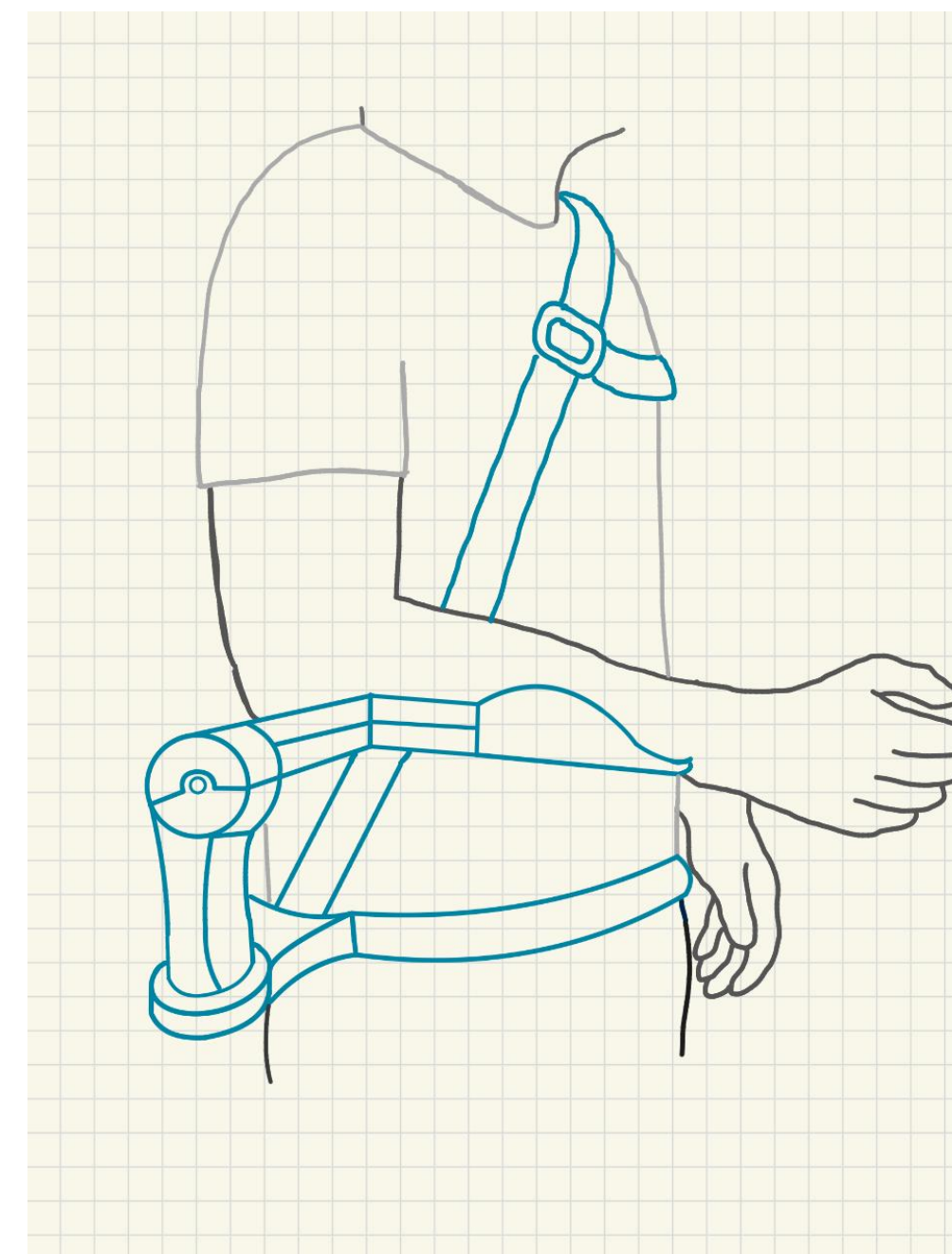
When: 2025-2026 school year

Project Milestones

Going into this year our goal was to design and create an assistive arthritis device for Christina to help relieve pain in her elbow when going about daily tasks.

Preliminary Design

- Hung off waist & shoulder
 - Too bulky
 - Some motion restricted
 - Uncomfortable
 - Telescoping Mechanism
- Single torsion spring

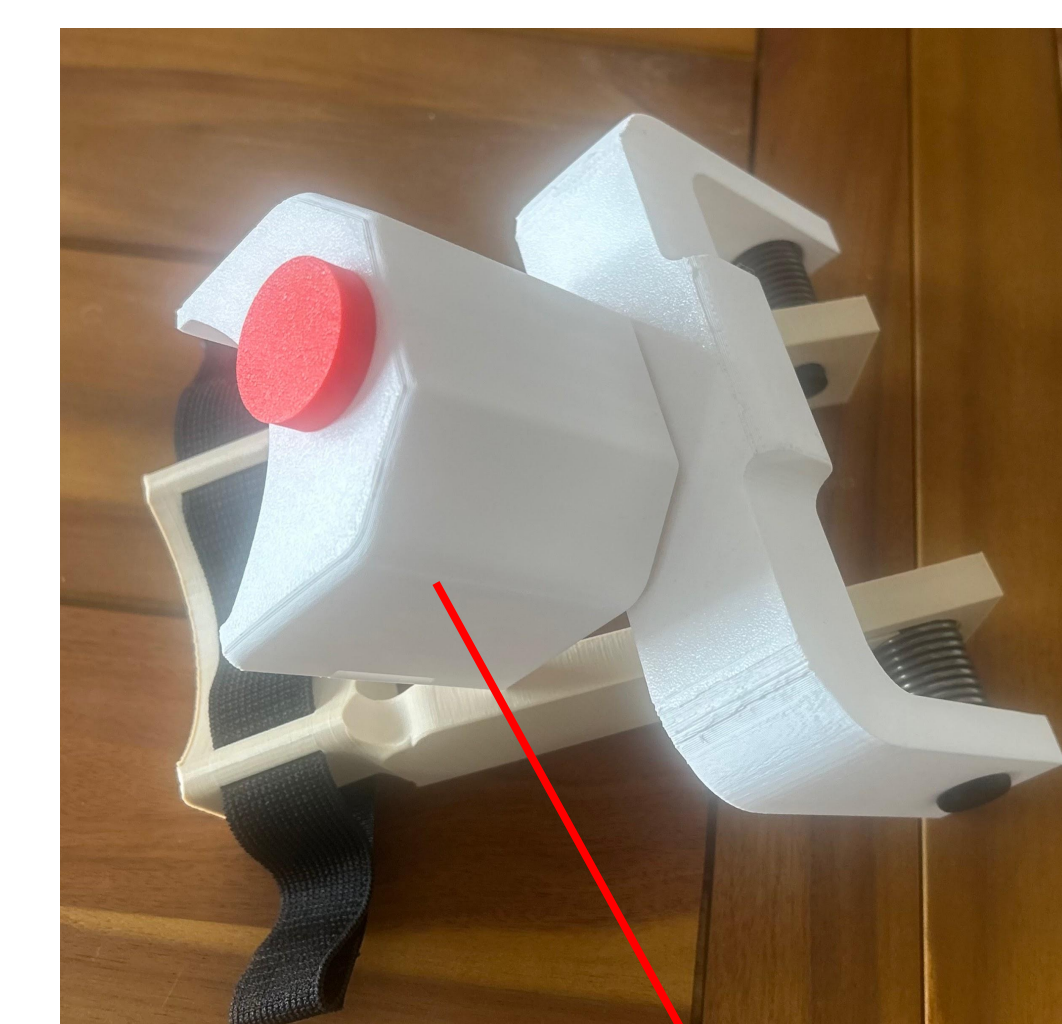


Current Design

- Attached only to arm
 - Compact / Portable
 - Simple to use
- Relieves elbow directly
 - Full range of motion
 - 60 Degree wrist rotation
- 2 Torsion springs



Torsion Spring



Bicep Support

Forearm Support

Manufacturing

Current Design:

- For manufacturing, we 3D printed our parts out of PETG
- We cut the legs of our torsion springs with a bolt cutter in order to fit our design

Future Considerations:

- We are considering adjusting to TPU for its rubbery texture and ductility.
- Attach a foam barrier between the cups and the bicep / forearm

Development & Testing

Development:

- Switched design from back-supported to arm-supported to minimize variables
- Developed spring insertions in the forearm and bicep-connections to house springs
- Using a pin and hole system to restrict excessive forearm rotation
- Planning on using TPU as our 3D printing material to maintain strength and have a ductile properties
- Adding Komprex foam padding and adjustable Velcro strap to the bicep support for comfort

Testing:

1. Ensure torsion spring is properly attached in housing
2. Flex the device, stretching the torsion spring
3. Wrap the Bicep strap first connecting device to bicep
4. Wrap the forearm cup around the forearm.
5. Adjust device if necessary
6. Interview user while they use the product.

System Overview

Our project had 3 separate components:

Bicep Connection:

- Attached to the bicep and extrudes to the elbow at an angle
- At the end of the extrusion, it houses the torsion spring.

Forearm connection:

- Houses the torsion spring
- Connected to the Bicep Connection via a rod
- Attaches to the forearm with a cup.

Torsion Spring:

- Connected to both the bicep and forearm connections.
- Located at the elbow, - its torque counterbalances the weight of the forearm.

Future Work

In the future, we hope to make our product more ergonomic to enhance comfortability. Additionally, we would like look into custom torsion springs to be compatible with Christina's exact measurements.

Thank you to our amazing team and the EMPOWER's board for their support. Also thanks to Christina for her advice and partnership!

Acknowledgements

